SE LAB 1

Requirements Engineering & UML Use-Case Modelling

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QUESTION NO. 26

Peer-to-Peer Carpooling & Expense Splitter

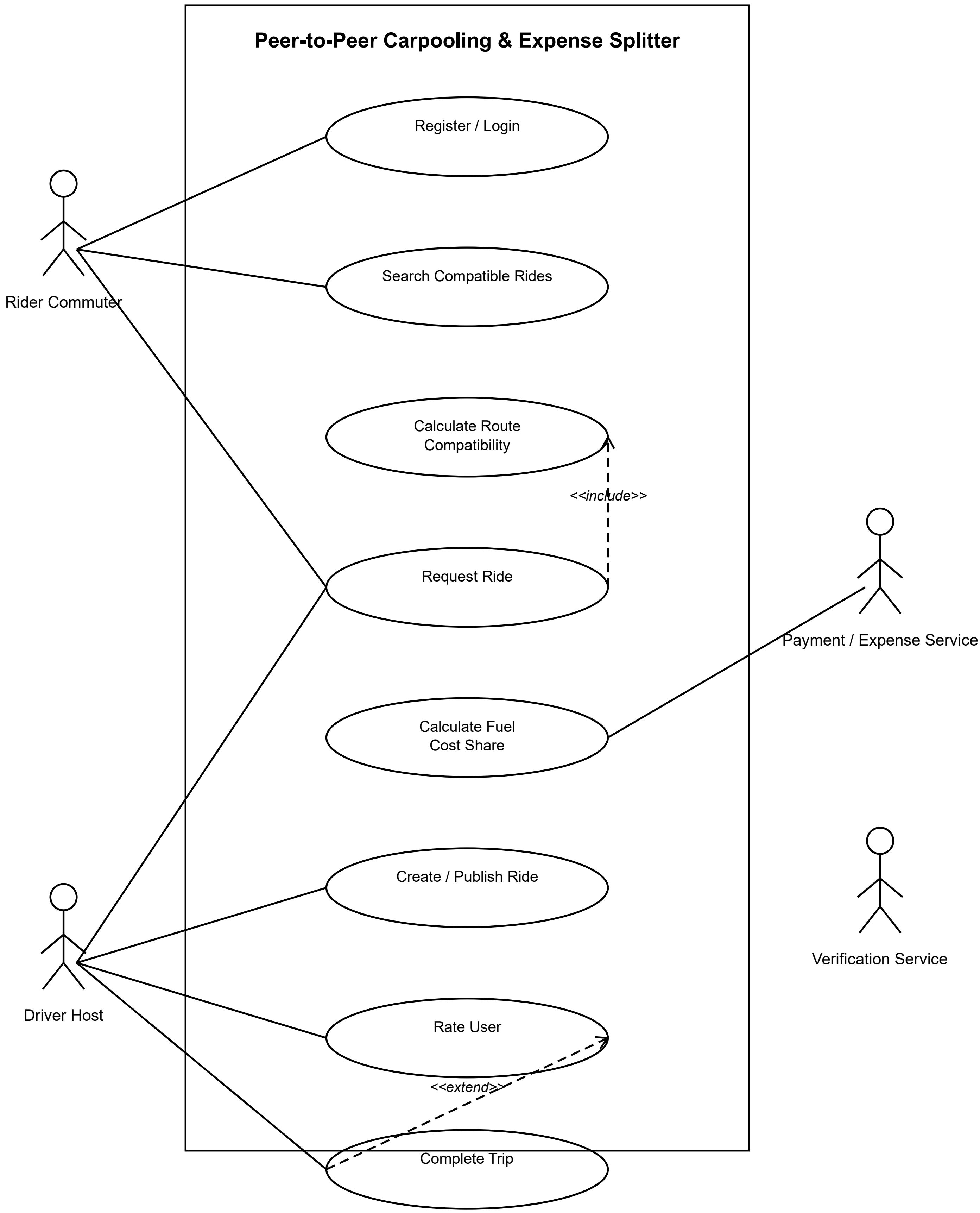
Functional Requirements (FR)

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall calculate route compatibility between driver and passenger commute paths, permitting matches only within a 1.5 km detour tolerance.	High	Pass: Route match is confirmed when the detour is ≤ 1.5 km. Fail: A match is accepted when the detour exceeds 1.5 km.	Ensures that matched rides remain within the permitted detour tolerance.
FR-002	Functional	The system shall allow a Driver Host to create and publish a carpool ride with route, date, time, seats, and ride details.	High	Pass: A valid ride is published and visible to eligible riders. Fail: An incomplete ride is published.	Allows drivers to make available rides discoverable to commuters.
FR-003	Functional	The system shall allow a Rider Commuter to search for and request a compatible carpool ride.	High	Pass: A rider can select a compatible ride and submit a request. Fail: An incompatible ride can be requested.	Provides the core ride-matching and request functionality.
FR-004	Functional	The system shall calculate and display the rider's automated fuel cost share for a matched trip.	High	Pass: A fuel-cost share is calculated and displayed before the rider confirms the request. Fail: No cost share is shown.	Provides transparent and automated expense splitting.
FR-005	Functional	The system shall allow Rider Commuters and Driver Hosts to submit ratings after a completed trip.	Medium	Pass: A rating is stored after the trip is marked completed. Fail: A rating cannot be submitted for an incomplete trip.	Supports accountability and trust between carpool participants.

Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Nonfunctional	User ratings, driving license verification documents, and travel histories must be stored securely with encrypted backups.	High	Pass: Security testing confirms encrypted storage and encrypted backups for the specified data. Fail: Any specified data is stored or backed up without encryption.	Protects sensitive user and travel information.
NFR-002	Nonfunctional	The system shall return route-matching and compatible-ride search results within 3 seconds under simulated peak load.	High	Pass: At least 95% of search requests return results within 3 seconds during peak-load testing. Fail: The target is not met.	Provides responsive matching during high-demand periods.

UML Use-Case Diagram — Question 26



Solid line = association   Dashed arrow = UML include/extend relationship

Use-Case Flow Specification

Use Case: Request Compatible Carpool Ride

Primary Actor: Rider Commuter

Supporting Actor: Driver Host

Goal: To allow a rider to find and request a suitable carpool ride based on route compatibility and the 1.5 km detour tolerance, with an automated fuel-cost share.

Preconditions

• The Rider Commuter is registered and logged in.

• The Driver Host has an active published ride.

• The rider has entered a pickup location and destination.

• The system can access route information for compatibility checking.

Postconditions

• The compatible ride request is recorded in the system.

• The Driver Host is notified of the request.

• The automated fuel-cost share is calculated and displayed.

• The ride request status is updated for the rider.

Main Success Scenario

1. Rider Commuter logs in to the system.
2. System authenticates the rider and displays the carpool search interface.
3. Rider enters the pickup location and destination.
4. System searches available rides published by Driver Hosts.
5. System calculates route compatibility between the rider's route and each driver's route.
6. System checks that the required detour does not exceed 1.5 km.
7. System displays the compatible rides to the rider.
8. Rider selects a suitable ride.
9. System calculates and displays the rider's automated fuel-cost share.
10. Rider confirms the ride request.
11. System records the request and notifies the Driver Host.
12. System displays confirmation of the successful request.
13. Use case ends successfully.

Alternate Flow – Detour Exceeds Tolerance

A1. At Step 6, the system determines that the required detour is greater than 1.5 km.

A2. System rejects that route as incompatible and does not allow the rider to request it.

A3. System informs the rider that the ride exceeds the permitted detour tolerance.

A4. Rider is shown other compatible rides, if available.

A5. If no compatible ride exists, the request ends without being submitted.